

AI Chatbot Specification

■ Clinics / Healthcare

AI chatbot specification for GP clinics, dental clinics, TCM, physiotherapy, and specialist practices in Singapore.

Suggested Pricing

Component	Price
One-time setup fee	\$500 - \$800
Monthly maintenance	\$100 - \$150/month
Add-on: WhatsApp integration	+\$150-200
Add-on: Payment collection	+\$100-150
Add-on: Custom integrations	+\$100-200 each

Business Pain Points (What You Sell Against)

These are the problems the business owner already knows they have. Lead with these in your pitch — don't sell 'AI', sell the solution to their pain.

- Phone lines jammed during morning peak — patients can't get through to book.
- Receptionist spends 60% of time answering same questions: 'Do you accept Medisave?', 'What are your rates?'
- No-shows waste doctor's time — especially for specialist appointments worth \$100-300.
- After-hours enquiries go to voicemail — patients call competitors instead.
- New patients don't know what to bring for their first visit — arrive unprepared, wasting time.
- Online reviews complain about 'hard to book' and 'long phone wait'.

30-Second Sales Pitch

I'll build an AI assistant for your clinic that handles appointment bookings, answers patient questions about services and pricing, and sends automated reminders — 24/7. Patients book when it's convenient for them, your receptionist handles in-clinic work instead of phone calls. PDPA-compliant, takes 4-5 days to set up.

Feature Requirements

Build in order: Must Have first (your MVP), then Should Have (first upsell), then Nice to Have (premium tier).

Feature	Description	Priority	Effort
Service & Pricing FAQ	Answer questions about services, consultation fees, insurance, Medisave	Must Have	Low
Appointment Booking	Book appointment with doctor, check availability, collect patient info	Must Have	Medium
Appointment Reminders	Automated reminders with what to bring and clinic directions	Must Have	Medium
Pre-visit Digital Forms	Send intake forms digitally before appointment to save time	Should Have	Medium
Insurance/Medisave Checker	Quick lookup if a procedure is claimable under patient's insurance or Medisave	Nice to Have	Medium
Queue Status	Real-time queue estimate for walk-in clinics	Nice to Have	High

Detailed Feature Specifications

Use these specifications when building with Claude Code. Each feature includes the user story, technical implementation notes, and acceptance criteria.

Feature: Service & Pricing FAQ

Priority: Must Have | **Effort:** Low

User Story: As a patient, I want to know how much a consultation costs and whether my insurance/Medisave is accepted.

Technical Notes: Upload fee schedule, accepted insurance list, Medisave-claimable procedures. Handle common queries: 'do you accept Medisave', 'how much for dental cleaning', 'do you take AIA insurance'. IMPORTANT: Bot must not provide medical advice.

Acceptance Criteria:

- Returns correct consultation fees
- Lists accepted insurance providers accurately
- States which procedures are Medisave-claimable
- NEVER provides medical advice or diagnosis suggestions
- Responds with 'Please consult a doctor' for medical questions

Feature: Appointment Booking

Priority: Must Have | **Effort:** Medium

User Story: As a patient, I want to book an appointment online without calling during busy hours.

Technical Notes: Flow: select service type -> select doctor (if multiple) -> show available slots -> collect patient name, NRIC (last 4 digits only for PDPA), phone, reason for visit. Create calendar event. CRITICAL: Do not store full NRIC or sensitive medical details in chat logs.

Acceptance Criteria:

- Shows real-time doctor availability
- Collects only necessary patient info (name, phone, last 4 of NRIC)
- Does NOT store full NRIC or sensitive medical info
- Sends confirmation with clinic address and what to bring
- Notifies clinic staff of new booking

Feature: Appointment Reminders

Priority: Must Have | **Effort:** Medium

User Story: As a clinic owner, I want patients reminded about their appointments to reduce no-shows.

Technical Notes: Send reminder 24hr before via WhatsApp/SMS. Include: date, time, doctor name, clinic address, what to bring (NRIC, insurance card, referral letter if applicable). Option to confirm or reschedule.

Acceptance Criteria:

- Sends reminder 24hr before appointment
- Includes 'what to bring' checklist based on appointment type
- Patient can confirm or reschedule via reply
- No-show rate tracked in admin dashboard

Feature: Pre-visit Digital Forms

Priority: Should Have | **Effort:** Medium

User Story: As a patient, I want to fill in my details before arriving so I don't wait at reception.

Technical Notes: Send link to digital form (Typeform/Google Form) 24hr before appointment. Collect: personal details, medical history, allergies, current medications, reason for visit. Auto-attach to patient's appointment record.

Acceptance Criteria:

- Form sent automatically 24hr before first visit
- Covers essential medical history safely
- Data stored securely (encrypted at rest)
- PDPA consent checkbox included in form

Feature: Insurance/Medisave Checker

Priority: Nice to Have | **Effort:** Medium

User Story: As a patient, I want to quickly check if my procedure is covered by Medisave or my insurance plan.

Technical Notes: Maintain a lookup table: procedure x insurance_provider x claimable (yes/no/partial). Data provided by clinic. Caveat: 'This is a general guide. Final claimability depends on your specific plan. Please bring your insurance card for verification.'

Acceptance Criteria:

- Returns correct claimability for known procedure+insurance combinations
- Always includes disclaimer about verifying with insurer
- Says 'I'm not sure' rather than guessing for unknown combinations

Feature: Queue Status

Priority: Nice to Have | **Effort:** High

User Story: As a walk-in patient, I want to know how long the current wait is before I travel to the clinic.

Technical Notes: Requires integration with clinic's queue management system or manual update by receptionist. Provide estimated wait time. This is complex — only offer if clinic has an existing queue system with API access.

Acceptance Criteria:

- Shows current estimated wait time

- Updated in real-time or within 10 minutes
- Includes disclaimer: 'Wait times are estimates and may vary'

Sample Conversations

These demonstrate expected chatbot behaviour. Use these as test cases during development.

Scenario: New patient booking a dental appointment

Customer: I need to see a dentist for tooth pain

Bot: I'm sorry to hear that! Dr. Lim has availability tomorrow at 10am or 3pm. For a dental consultation, the fee is \$50-80 (Medisave claimable for certain procedures). Which time works for you?

Customer: 10am

Bot: Great. Can I get your name and phone number to confirm the booking?

Customer: John Lee, 9111 2222

Bot: Booked! Dr. Lim, tomorrow 10am. Our clinic is at 456 Tampines St 42, #01-10 (near Tampines MRT). Please bring your NRIC and insurance card. You'll receive a reminder tonight. Get well soon!

Scenario: Insurance enquiry

Customer: Do you accept Great Eastern insurance?

Bot: Yes, we accept Great Eastern insurance for most consultation and treatment services. The coverage and claimable amount depends on your specific plan. Please bring your insurance card when you visit, and our staff will verify your coverage at the reception. Would you like to book an appointment?

Technical Architecture

Core Stack (Required)

Component	Technology	Purpose
LLM	Claude API (claude-sonnet-4-20250514)	Understands questions and generates answers from context
Vector Database	ChromaDB (local) or Pinecone (cloud)	Stores document embeddings for RAG retrieval
Embedding Model	Voyage AI or OpenAI text-embedding-3-small	Converts text chunks to vectors
Backend	Python FastAPI	API server handling chat requests
Frontend Widget	Vanilla JS embeddable snippet	Chat bubble on client's website
Admin Dashboard	Streamlit or Next.js	Where business owner uploads docs and views analytics

Add-on Integrations

- **Google Calendar API:** Doctor availability, appointment scheduling
- **WhatsApp Business API:** Appointment confirmations, reminders, patient communication
- **Typeform / Google Forms:** Pre-visit intake forms
- **Telegram Bot:** Staff notifications for new bookings
- **Stripe / PayNow:** Deposit collection for specialist appointments to reduce no-shows

Data Model (Documents to Collect from Client)

These are the documents and data you need from the business owner during onboarding. The quality of these documents directly determines chatbot accuracy.

- **Fee Schedule:** All services: consultation fee, procedure fees, itemised pricing
- **Insurance & Medisave Guide:** Accepted insurers, Medisave-claimable procedures, claim process
- **Doctor Profiles:** Doctor names, specialisations, qualifications, available days
- **Clinic Info:** Address, operating hours, contact, directions, parking
- **First Visit Checklist:** What to bring for each type of appointment
- **Clinic Policies:** Cancellation policy, no-show policy, walk-in vs appointment policy

PDPA Compliance Checklist (Singapore)

- Display clear data collection notice before collecting any personal data
- Get explicit consent before storing name, phone, email
- Provide opt-out option: 'Type STOP to opt out of data collection'
- Do not store NRIC, credit card numbers, or medical records in chat logs
- Implement data retention policy: auto-delete chat logs after 90 days
- Provide data export/deletion on request from the business owner
- Add privacy policy link in the chat widget footer

Singapore Localisation Notes

- Support Singlish phrasing: 'got discount ah?', 'how much one?', 'can book or not'
- Handle mixed language queries (English + Mandarin terms are common)
- Use SGD (\$) for all pricing — never USD
- Time format: 12-hour with AM/PM (e.g., 3pm, not 15:00)
- Support PayNow QR code generation for payments
- Reference local landmarks for directions ('near Jurong East MRT')
- Public holiday awareness for Singapore calendar